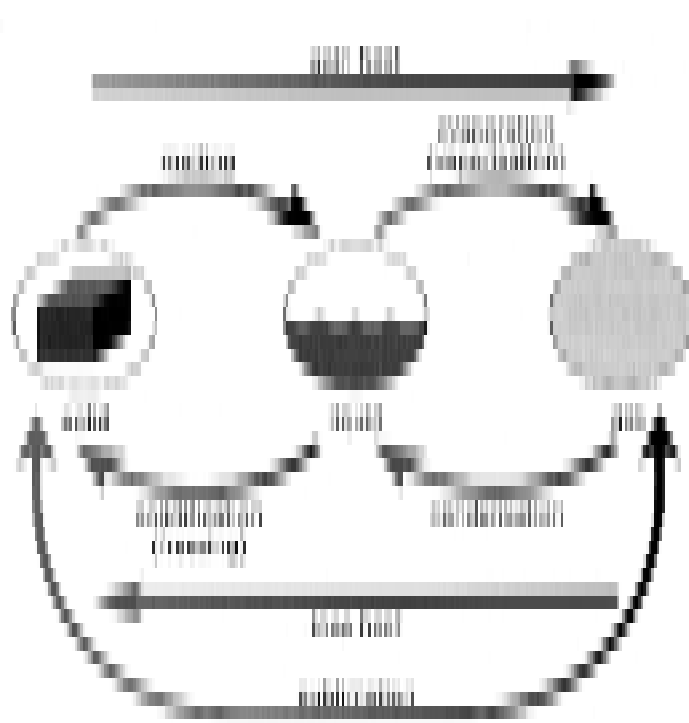


CHANGING STATES



MELTING

- Change from solid to liquid.
- Particles gain enough energy to break away from their fixed positions.

BOILING

- Change from liquid to gas.
- Particles gain enough energy to overcome any attraction for other particles.

CONDENSING

- Change from gas to liquid.
- Particles lose heat energy and move closer together.
- Attractive forces pull them together.

FREEZING

- Change from liquid to solid.
- Particles lose energy, move closer together and attractive forces hold them in fixed positions.

SUBLIMATION

- Change from solid to gas (or gas to solid) without a liquid phase.

e.g. Dry ice (solid carbon dioxide) goes straight to a gas, hence it is “dry” (no liquid).

EVAPORATION

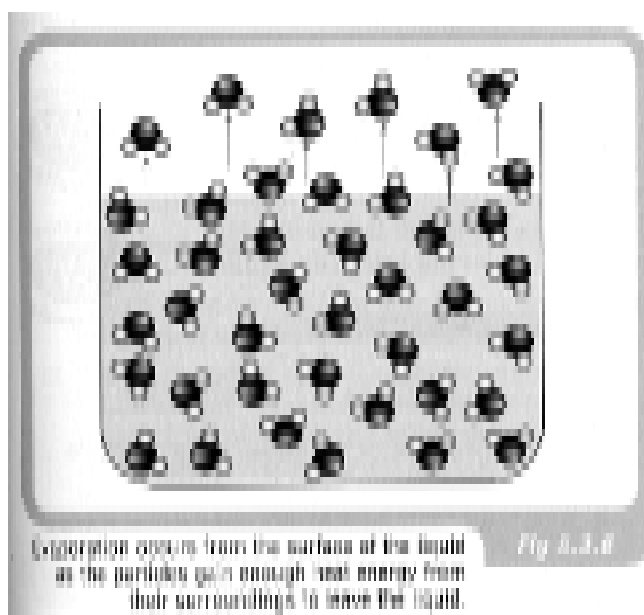
- Change from liquid to gas at a temperature below the boiling point.
- Occurs from the surface of a liquid.
(Compare with boiling which occurs at the boiling point only and from within the liquid.)

If the attractions among the particles of a substance are **weak**, the substance is a gas at room temperature.

e.g. Oxygen and carbon dioxide are gases.

If the attractions are **strong**, the substance is a liquid or solid at room temperature.

e.g. Mercury is a liquid metal, steel is a solid metal.



CHANGES OF STATE AND TEMPERATURE

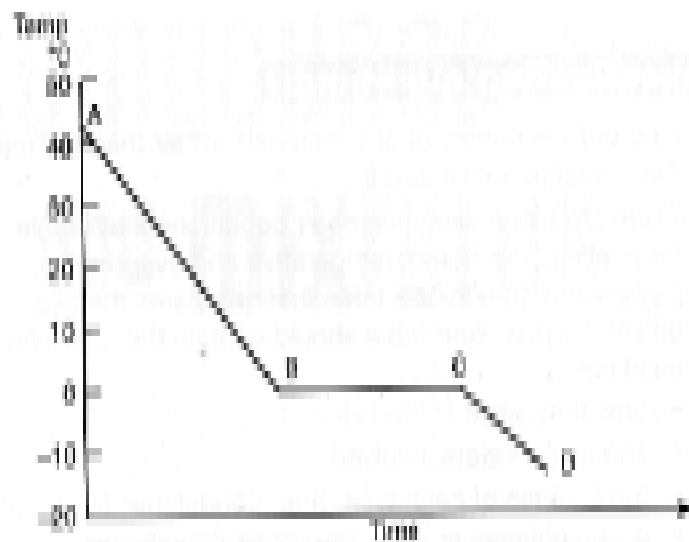
When a substance changes phase, *no temperature change occurs* to it.

Instead, heat is being used to *change the distance between the particles* (either closer together (freezing or condensing) or further apart (melting or boiling).

This heat is called *latent heat* (meaning "hidden heat") as no temperature change is measured.

Example

Consider the graph below which shows a cooling curve for water initially at about 40 °C.



Section AB: shows the water cooling as heat is removed.

Section BC: water freezes as heat is removed and particles move closer together.

Section CD: ice continues to lose heat and drops below zero.